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Review

The modifying factors that help improve anterior cruciate ligament reconstruction rehabilitation: A narrative review



Alexandre JM Rambaud^{a,b,c,*}, Thomas Neri^{a,d}, Bart Dingenen^e, David Parker^{f,g}, Elvire Servien^{h,i},
 Alli Gokeler^j, Pascal Edouard^{a,k}

^a Univ Lyon, UJM-Saint-Etienne, Laboratoire Interuniversitaire de Biologie de la Motricité, EA 7424, F-42023 Saint-Etienne, France

^b Motion Analysis Lab, Physiotherapy and Sports Medicine Department, Swiss Olympic Medical Center, La Tour Hospital, Meyrin, Switzerland

^c SFMKS-Lab, Société Française des Masseurs-kinésithérapeutes du Sport, Pierrefitte/Seine, France

^d Department of Orthopaedic Surgery, University Hospital of Saint-Etienne, Faculty of medicine Saint-Etienne, France

^e Reval Rehabilitation Research Center, Faculty of Rehabilitation Sciences, Hasselt University, Agoralaan A, 3590 Diepenbeek, Belgium

^f Sydney Orthopaedic Research Institute, Sydney, Australia

^g The University of Sydney, Sydney, Australia

^h Univ Lyon, Laboratoire Interuniversitaire de Biologie de la Motricité, EA 7424, F-42023 Lyon, France

ⁱ Department of orthopaedic surgery and sports medicine, FIFA medical center of excellence, Lyon University Hospital, Lyon, France

^j Exercise Science & Neuroscience Unit, Department Exercise & Health, Faculty of Science, Paderborn University, Paderborn, Germany

^k Department of Clinical and Exercise Physiology, Sports Medicine Unity, University Hospital of Saint-Etienne, Faculty of medicine, Saint-Etienne, France

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ABSTRACT

Background: : The goal of a rehabilitation programme after anterior cruciate ligament (ACL) reconstruction is to manage a patient's goals and expectations (i.e., returning to physical activities and sports) while minimizing the risk of new injury, particularly a new ACL injury. Although general rehabilitation programmes have been proposed, some factors can lead to adapting each programme to each patient.

Objective: : To describe how different variables, including surgical techniques, sports participation, psycho-social and contextual factors can modify the rehabilitation programme.

Methods: : We performed a narrative review with input from experts in the field (level of evidence 5).

Conclusions: : Modifying factors of the ACL rehabilitation programme are related to the initial lesion or surgery, to sports, or to psychological or social aspects. Regarding the type of graft, the rehabilitation is mainly different in the early postoperative phase; the other phases are not graft-based but rather goal-based rehabilitation. Depending on the meniscal or cartilage repair, the rehabilitation protocol will initially take priority over the anterior cruciate ligament reconstruction protocol. The ACL reconstruction rehabilitation programme should meet the requirements of the anticipated sports, to optimize the athlete's ability to return to the expected level and minimize the risk of reinjury. Psycho-social and contextual factors must also be considered in rehabilitation care to individualize and optimize each patient's programme.

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Introduction

Rehabilitation after anterior cruciate ligament reconstruction (ALCR) is a long and challenging process for both the patient and health professionals. The goal of the rehabilitation programme is to recover function for daily life, including work and participation in physical activities and sports. The ultimate goal of management is to achieve the patient's objective of returning to their previous level of

sport, if applicable, while minimizing the risk of new injury, particularly a new anterior cruciate ligament (ACL) injury that leads to high healthcare costs and increased disability. **Unfortunately, for the athletes, the rate of return to sport (RTS) at the same level after 3 postoperative years is only 63% [1] and there is a major risk of new injury (20%), especially to the contralateral knee (12%) in patients returning to sport [2].**

To facilitate this process, Andrade et al. [3] proposed a synthesis of various clinical practice guidelines regarding rehabilitation after ALCR. While these recommendations may be useful as a general guide, all patients are different. Therefore, we think that this standardised rehabilitation programme should serve as a basis and be modified and adapted to the patient's individual aspects, such as the ACL injury and its consequences on stability, in addition to potential

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* Corresponding author at: Université de Lyon, UJM-Saint-Etienne, Laboratoire Inter-Universitaire de Biologie de la Motricité, EA 7424, Campus Santé Innovations IRMIS 42270 Saint-Priest en Jarez, France.

E-mail address: alexandre.rambaud@univ-st-etienne.fr (A.J. Rambaud).

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associated lesions, sports participation, professional activities, psycho-social and contextual factors of the patient, which we refer to as “modifying factors”.

These modifying factors can be intrinsic factors (corresponding to the individual patient’s characteristics) or external factors (depend on the external environment or extrinsic factors) (Fig. 1). They illustrate the complexity of managing the rehabilitation of patients with ACLR for health professionals. Therefore, the aim of this article is to describe these modifying factors with a narrative review and inputs from experts in the topic, to help individualize the ACLR rehabilitation programme.

Do surgical factors change the ACLR rehabilitation management?

Effect of ACLR graft type

General information on ACLR

Several types of grafts are used for ACLR after primary ACL injury. The most common reconstruction technique uses the hamstring tendon autograft (semitendinosus) [4]. Other commonly used techniques use extensor mechanism autografts: the bone patellar tendon bone (BPTB) with patellar and tibial bone plugs or the quadriceps tendon [5]. Allograft and synthetic grafts will not be discussed here because they are less common and are reserved for specific cases. Because ACLR rehabilitation must include restoration of knee function as well as healing of the donor graft site, the rehabilitation programme must be first tailored to the type of graft used. Despite no clinical difference in terms of functional outcomes [6] and no major differences in re-rupture [7], each technique presents specific complications that need to be considered during rehabilitation (Appendix [8].

For both techniques, postoperative rehabilitation should focus on decreasing knee swelling and pain, then on increasing passive knee extension. It must also be focused on voluntary quadriceps contraction during the postoperative phase and throughout rehabilitation to counteract the onset of arthrogenic muscle inhibition [9] and to guide

the healing of the graft donor site. Open and closed kinetic chain can be used for quadriceps and hamstring strengthening, but the optimal timing for the start, range of motion, and applied resistance for open kinetic chain after ACLR is debated [10]. General complications related to arthroscopy and more specific complications related to the surgical reconstruction technique have been reported and are listed in the Appendix.

ACLR with hamstring tendon graft

Following ACLR with a hamstring tendon graft, the remaining muscle stumps of tenotomized semitendinosus may be the site of a “lizard tail phenomenon” [11] and could lead to regrowth of a neo-tendon. A local hematoma is formed at the transplant donor site, which may result in bruising of varying degrees. This haematoma seems to play a role by leading to the proliferation of fibroblasts and collagen production in the harvesting defect, allowing for the initiation of the regeneration process [12].

A neo-tendon re-forms at the donor site in most patients with hamstring graft, but the process is long (~18 months) and may not occur at all in 10% to 50% of patients [13]. The other hamstring muscles, via hypertrophy, would have to then compensate for a loss of strength in the semitendinosus muscle owing to the absence of the distal tendon. However, Xergia et al. [14], in a systematic review and meta-analysis, reported a loss of strength in the knee flexors 2 years after ACLR with semitendinosus tendon as compared with BPTB graft. Recent results confirm the impact of a specific strength training during rehabilitation (based on the American College of Sports Medicine principles and previous studies by the authors) with very good results on the strength of the quadriceps and hamstring muscles at 7 months (i.e., no significant differences in strength as compared with a control group) [15], which highlights that an optimized rehabilitation is possible and effective. Optimizing the recovery of hamstring muscle strength is a major goal for clinicians because this muscle plays an essential role in protecting the ACL and therefore the neo-ligament.

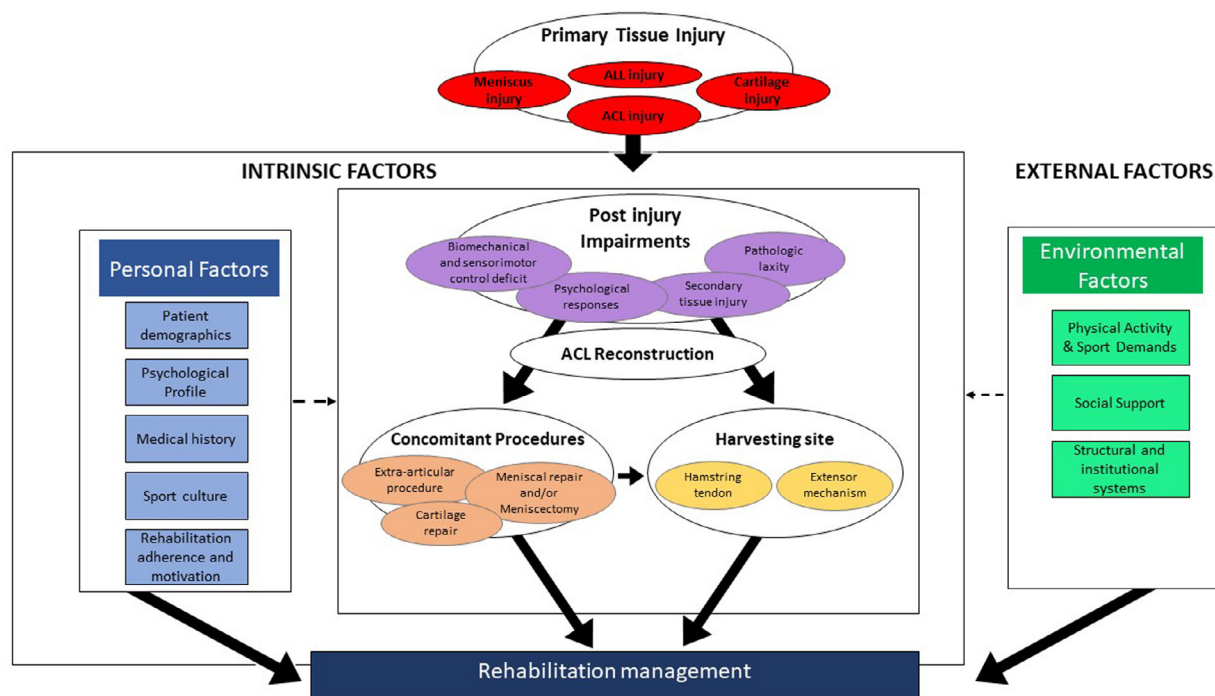


Fig. 1. Modifying factors after anterior cruciate ligament (ACL) reconstruction inspired by Hertel al. (2019). Intrinsic factors correspond to the individual athlete’s characteristics (personal factors, self-driven characteristics) and external factors to factors that patients were not able to actively control or an external environment. ALL, anterior lateral ligament

ACLR with BPTB graft

Following ACLR with BPTB, as with a hamstring tendon graft, the goal of postoperative rehabilitation is full extension of the knee, but the donor harvesting site on the extensor mechanism requires special attention to the voluntary contraction of the quadriceps. It is not uncommon to find persistent anterior knee pain, kneeling pain [15], and arthrogenic muscle inhibition may also contribute to poor quadriceps muscle strength recovery [9]. Hamstring muscle strengthening can be performed immediately and can be done with open kinetic chain exercises, depending on recovery of range of motion of the operated knee.

For the strengthening and neuromuscular rehabilitation phase, the exercises are the same as for a hamstring graft. The specific exercises for strengthening are important, and quadriceps open kinetic chain exercises [10] must be associated with the other exercises in closed kinetic chain recommended for the gain of muscular strength [15]. Special attention should be placed on the presence of anterior knee pain, a common complication of BPTB ACL reconstruction [16]. Restoring full knee hyperextension is essential to reduce anterior knee pain (with any type of graft) [17]. As has been reported for strength deficit after hamstring graft, a strength deficit of the muscle involved by donor site (quadriceps muscle) has been reported for BPTB grafts [14]. For quadriceps tendon graft, the clinical and functional outcomes seem comparable to those with BPTB, with less graft donor site pain [5].

In conclusion, regardless of the type of graft, the rehabilitation mainly differs in the early postoperative phase. The other phases are not a graft-based but rather a goal-based rehabilitation.

Effect of extra-articular procedures combined with ACL reconstruction

In case of ACLR with a lateral extra-articular procedure (see [18–27]), the instructions for postoperative rehabilitation are essentially the same as for isolated ACLR. In early postoperative care, no specific additional bracing or unloading is required. There is no evidence to indicate whether the addition of an anterolateral procedure to ACLR affects the postoperative range of motion [19,27]. Consequently, patients with combined procedures may follow a routine rehabilitation protocol aimed at controlled restoration of range of motion, muscle strength and neuro-muscular control. The only difference compared to isolated ACLR is the additional wound on the lateral side of the knee but without any reported functional issues and no need for additional care [19,27]. The rehabilitation programme may be started immediately to help gradually restore mobility and resume usual daily activities.

To date, the addition of an extra-articular procedure has been found effective in reducing the re-rupture rate [28,29], with a lower rate of failure of medial meniscal repairs [30] and a higher rate of patients returning to their previous level of sports performance [28,29], but no studies have been able to demonstrate that it reduces the time to RTS. As compared with isolated ACLR, the addition of a lateral procedure is not associated with increased surgical complications [25], not limiting the RTS (for any type of sport), and is recommended in pivoting sports in high-level competitive athletes and those considered at high risk of reinjury [20,29].

Do concomitant lesions change the ACLR rehabilitation management?

Special considerations for rehabilitation after ACLR with meniscal pathology

After ACLR combined with meniscal surgery, the protocol needs to combine the principles of ACLR and meniscal surgery rehabilitation, without compromising the recovery of either pathology. In most cases, the addition of meniscal pathology has no direct implications for rehabilitation, and the general principles of ACLR rehabilitation can be followed with possibly some minor variations. The literature

lacks clear consensus regarding rehabilitation protocols, particularly with combined ACLR and meniscus surgery, and there is considerable variation of individual surgeon preferences [31,32].

In patients with additional meniscectomy at the time of ACLR, usual rehabilitation protocols can be followed, but each protocol may need to be individualized depending on the nature of the meniscectomy. In particular, the impact of the meniscectomy on the loading of the affected compartment may have implications for how the knee responds when loads are reintroduced. If a small percentage of a meniscus has been removed, and there is a stable intact rim, loading of the compartment will be modestly affected and the usual rehabilitation can be followed. In cases with more significant meniscectomy, particularly those with resection closer to the rim, and more so with lateral meniscectomy, the increased loading may manifest as pain and swelling, with an increased risk of chondral wear. In these patients, ensuring that pain and swelling are completely resolved would be sensible before starting activities requiring higher levels of loading and impact on the joint, such as running, jumping, plyometric and side-cutting activities. The introduction of such activities should be performed gradually, with monitoring for the onset of pain or reactive joint effusion and subsequent modification of loads to manage this.

In cases that involve meniscal repair, the modifications to the rehabilitation programme will vary depending on the nature of the repair. In cases in which a peripheral longitudinal tear has been repaired, there is good evidence that weight bearing and progressive range of motion do not cause displacement of the tear or compromise healing [33,34]. For this reason, although some surgeons may limit flexion to approximately 90 degrees for 4 weeks, there is good evidence that allowing patients to progress to weight bearing and range of motion as tolerated, in line with a standard ACLR rehabilitation programme, is appropriate. In all cases of meniscal repair, avoiding any forced passive flexion for a minimum of 3 months would be sensible, particularly with weight bearing, but allowing patients to progress with active flexion as tolerated.

Less commonly, the healing of certain patterns of meniscal repairs may be compromised by weight bearing, and in these cases, limiting weight bearing for the initial phase of rehabilitation may be appropriate. Specific examples are meniscal root repairs [35] and repairs of large radial tears. In this type of tear, the compressive force of weight bearing may lead to distraction and failure of the repair, and therefore the meniscal repair rehabilitation protocol will initially take priority over the ACLR protocol. While specific details of each operation need to be considered when prescribing rehabilitation protocols, as a general guide in this type of repair, restricting the patient to touch weight bearing for 6 weeks and progressing gradually thereafter would be appropriate. There is less evidence regarding motion restriction, but to reduce the posterior meniscal translation that occurs in deeper flexion, limiting flexion to 90 degrees for the first 4 weeks and progressing gradually thereafter would be sensible as well as avoiding isolated hamstring activation for 6 weeks.

Cartilage repair combined with ACLR

At the time of ACLR, cartilage injuries are often encountered [36–41]. Meniscal lesions associated with ACL tears are well explored and described in literature, but there are few studies focused on cartilage damage and the associated mid- and long-term consequences [42]. Reported incidence of chondral lesions with an ACL tear is 16% to 46% [36–41]. Cartilage lesions usually occur at the time of the trauma, and should be taken into account because they may affect the rehabilitation programme and RTS. ACLR associated with cartilage damage leads to poorer outcomes as compared with isolated ACLR [41,43] and to increased risk of osteoarthritis at long-term follow-up.

The specific cartilage repair technique used will be critical to design the rehabilitation programme because different procedures

require specific programmes. Microfracture and microperforations are indicated in the treatment of International Cartilage Repair Society III-IV lesions, and allow full weight bearing during the postoperative period [44,45]. Procedures such as mosaicplasty [46], autologous chondrocyte implantation, or osteochondral allograft only require delayed weight bearing for 45 to 60 days. Within these restrictions, range of motion and muscle strengthening can be progressed as for isolated ACLR.

After recovery to full weight bearing, recommendations will not differ from isolated ACLR. Gaweda et al. [47] found slower recovery in patients who underwent ACLR and mosaicplasty at 6 months but no significant difference at 1 year postoperatively. According to Spike et al. [48], a return to high-level sports may be unrealistic with ACLR and autologous chondrocytes implantation.

At long-term follow-up, cartilage damage associated with ACLR has led to a significant decrease in function scores [49–51]. However, Ulstein et al. [52] showed different results, with no negative effect on patient-reported outcome 5 to 9 years after associated ACLR and cartilage damage management (radiofrequency debridement or microfractures). Some authors [44] suggest that cartilage status may be more important than meniscal lesions at the time of the surgery. Cartilage damage must be taken into account at the time of surgery, and all procedures can usually be performed at the same time as ACLR.

Effect of sports on the rehabilitation programme after ACLR

The desire of the athlete to return to a particular sport after ACLR should be reflected in the rehabilitation programme, such that sport-specific characteristics are included. The aim of the rehabilitation is to achieve physical and psychological readiness to RTS, and therefore the requirements of the rehabilitation will vary according to the type of sport (i.e., contact vs non-contact, pivoting vs non-pivoting) and the level of sport participation (non-elite vs elite athletes). Rather than a one-size-fits-all approach, the rehabilitation programme should be individualized based on the goals of the patient, reflecting the physiological and psychological requirements of the relevant sports the athlete expects to return to [53].

Pivoting versus non-pivoting and contact versus non-contact sports

Because ACLR rehabilitation should prepare athletes for participation in a desired sport, the rehabilitation will be different for pivoting and non-pivoting sports as well as contact and non-contact sports.

For non-pivoting sports, as soon as the patient masters exercises and their body in one axis and is able to increase the volume and intensity gradually, without provoking pain or effusion, a progressive programme [54] toward reaching the requirements of their specific sport can then ensure readiness for RTS while respecting the appropriate graft healing time. Thus, the ACLR rehabilitation programme and time for RTS for non-pivoting sports could be shorter than for patients who would like to return to pivoting sports. Although these patients should follow a complete ACLR rehabilitation programme and respect the usual healing graft and donor site timeline to limit the risk of re-injury, incorporating progressive exercises meeting the requirements of the sport activity may allow them to return to their sports earlier. For pivoting sports, the patient should master their body in the sagittal plane, progressively increase load, then master the movement in the frontal plane with a slow motion and increase load, and finally progressively move to combined plane and change of direction first with slow motion and low load to progressively increase the load to mimic that required by the sport.

For contact sports, the gradual increase in complexity of exercises should allow a patient first to master their body alone and then with other players. The ACLR rehabilitation programme should include such aspects to prepare the patient for a RTS.

Differences for the RTS with various sports have been reported. From a recent meta-analysis, the average time for RTS in elite soccer players was 6 to 10.2 months postoperatively, US football players 8.2 to 13 months and basketball players 10.7 to 11.8 months [55].

Level of sports

The level of practice will also affect the requirements of the ACLR rehabilitation programme. The general fitness of the patient/athlete should be restored to the pre-injury level (or more) in order to allow the training and competition workload encountered in sport. In addition, the knee and lower limb should ideally recover perfect function and physical capability, including endurance of strength and neuromuscular control, to assume these training and competitive loads [56]. If a patient/athlete normally trains 2 times a day for 5 days, plus one competition per week, the ACLR rehabilitation programme, helped by the physical conditioning programme, should gradually restore the patient/athlete to that load while respecting the knee, graft and donor site. A non-elite or non-professional athlete will not have the same requirements, and the rehabilitation should be adapted to the athlete's aim. Therefore, goals and expectations for the individual sporting participation should be clearly defined with the patient/athlete at the start of the ACLR rehabilitation programme. Also, an elite or professional athlete should have access to rehabilitation resources (e.g., professionals, structures, time to do rehabilitation).

Progressive increase of workload and sport-specificity toward a RTS

Recent studies [57,58] have reported that > 85% of patients do not meet the RTS criteria after 9 postoperative months. These results suggest that rehabilitation programmes are unable to prepare the patient adequately for RTS. Training loads may not be sufficient during rehabilitation, which could have significant effects when increasing the training load by returning to the field because an increase in training load could greatly increase the risk for re-injury [59].

Subsequently, Welling et al. [15] modified their previous strengthening programme. In a cohort of soccer players after a classical ACLR, without grade 3 or 4 meniscal and/or cartilaginous lesions, no quadriceps strength deficits were found when participants were tested at 7 months after ACLR as compared with a control group [15]. These findings suggest that the potential for physical recovery (e.g., strength, power, endurance) is not fully utilized in standard rehabilitation programmes.

Buckthorpe et al. [60] recently updated the RTS framework to strengthen a transition from rehabilitation to performance-based training and a continuum of on-field rehabilitation. On-field rehabilitation acts as a bridge between the gym-based rehabilitation environment and a return to the team. It involves a bespoke programme to simulate the sport-specific demands in line with the functional status of the player, to safely return to training. An effective on-field rehabilitation is characterized by a structured approach to planning and managing variation in training activity and load. Training acts as a stimulus for adaptation, and a progressive increase in the training load underpins an increase in the body's capacity to do work [61]. The training load can be progressed by changing the volume (the quantity of activity performed), intensity (the load per unit of time), and specificity (e.g., referring to the physiological, biomechanical, technical and psychological aspects of the sport) and frequency (the number of sessions in a period of time) of training in a way that is individualized to the athlete's capacity. Therefore, the workload should be increased progressively during the last part of rehabilitation by progressively adding more sports-specific training such as on-field training focused on agility and reactive training focused on tests, including fatigue conditions [62].

To conclude, previous capacities should be recovered, even sometimes to a greater extent than before injury. Indeed, at the time of injury, a deficit or asymmetry could exist, increasing the risk of the occurrence of injury [63]. Furthermore, the ACLR rehabilitation programme should meet the requirements of the sport, to optimize the chance of the patient/athlete returning to sport at their expected level and minimizing their risk of reinjury. There should be a gradual progression from the protected and controlled exercises performed initially in the clinic to the less protected and controlled, more functionally specific, and relevant activities of the sport, position, and style of play to which the athlete will be returning [64].

Psycho-social-contextual factors

Traditionally, most rehabilitation programmes mainly focus on the resolution of the physical impairments after ACLR. However, even if “normal” physical recovery is achieved, many athletes still fail to return to activity or sport performance. Psychological, social and contextual factors play a critical role in the recovery after ACLR. Therefore, a multifactorial biopsychosocial approach is warranted [65,66]. Their several factors, evaluation and intervention were presented in Fig. 2.

Psychological factors

An ACL injury leads to physical impairments and also has a psychological impact. Psychological factors surround cognitive, affective and behavioral responses are associated with an individual's experience of

injury, rehabilitation, surgery and RTS [66]. Common psychological responses are anxiety, depression, decreased self-esteem, confidence and fear of re-injury [67–69]. A systematic review revealed that 65% of patients cited a psychological reason for not returning to sport with fear of re-injury the most common reason [70]. Negative psychological responses may hinder rehabilitation, whereas positive psychological responses may facilitate recovery [71].

Social factors

Social factors include social networks and social exchanges that shape an individual's experience of injury, rehabilitation, surgery and RTS [66]. They include but are not limited to school or university coursework, work, family and entourage. A continued re-evaluation of social support needs is required to provide the amount and type of social support needed by the athlete recovering from an ACL injury [66] and after ACLR.

Contextual factors

Pain, performance level, and availability for sports participation are the main constructs to define a sports injury as perceived by elite athletes, coaches, and physiotherapists [72]. However, contextual factors can affect their perception and consequently their definition of a sports injury [72]. Contextual factors include structural and institutional systems in which an individual is embedded that influence their experience of injury, rehabilitation, surgery and RTS [66]. Themes such as environmental influences and sport culture can play

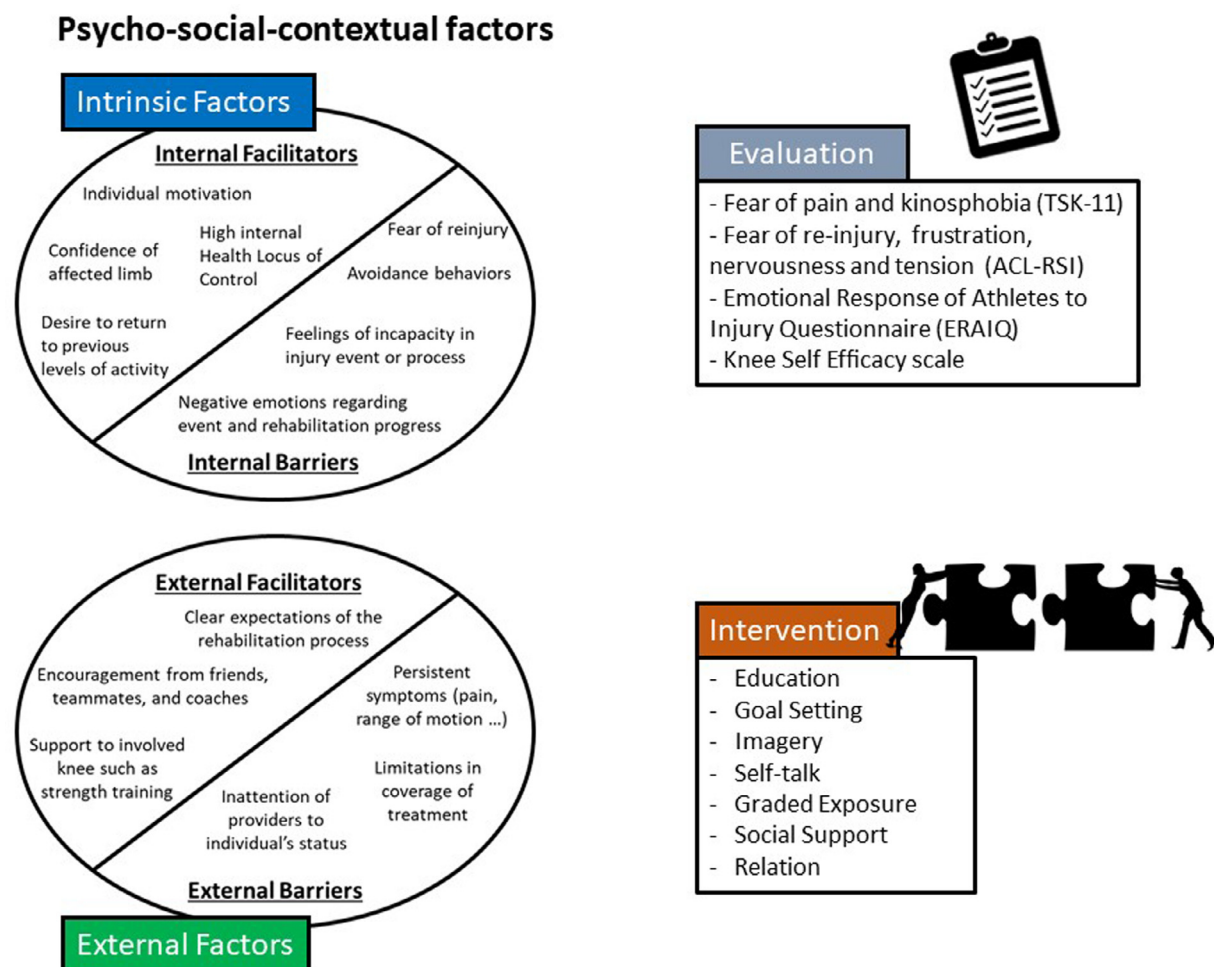


Fig. 2. Example of evaluations and interventions for psycho-social-contextual factors that can be broadly divided into 2 groups: intrinsic factors and external factors.

an important role because they determine and shape patient behaviours [66].

Implications

Clinicians must create an environment for the athlete that decreases negative responses and maximize rehabilitation adherence [73]. Incorporating a social support system and intrinsic motivation have been suggested as important factors for enhancing rehabilitation adherence and decreasing negative affective responses [73]. Effective communication (starting preoperatively) between patients and clinicians, building a strong therapist–patient relationship, shared decision-making, developing active coping and independence, patient and therapist engagement in care, strong social support, goal setting, positive self-talk, graded exposure, imagery and relaxation have been proposed to improve patient satisfaction, motivation and self-efficacy, leading to less unfulfilled patient expectations, fear of re-injury and improved RTS outcomes [66,69,71,74]. Most literature on psychological responses after knee injury lacks qualitative data from an athlete's perspective. From interviews with female football players after ACLR, 3 core themes representing psychosocial factors were found to enhance a positive coping style: 1) constructive communication and interaction with significant others; 2) self-efficacy; and 3) being able to set reasonable goals [75].

Assessment, monitoring and management of the psychological, social and contextual factors should be conducted throughout recovery (e.g., in Fig. 2), in addition to the physical aspects that need to be improved after ACLR [66]. Because these factors are dynamic and can change over time, this multifactorial approach is essential throughout the whole rehabilitation continuum and not just at the return to sport stage. These temporal aspects are highly variable among individuals. Therefore, individual consideration and management of these factors is needed [66].

Conclusion

There are as many different types of rehabilitation as there are patients. Despite some general guidelines for the rehabilitation programme, there are several elements that can influence it. The donor harvesting site mainly affects the immediate postoperative care and early phase of rehabilitation. No change in the rehabilitation programme is required if the reconstruction is combined with an extra-articular procedure. In the case of associated lesions requiring surgical repairs, the early phase may be modified according to the site of the lesion and its repair. Cartilage and meniscus injuries must be addressed concurrent with the ACLR protocol, and their specific rehabilitation protocol will often initially take precedence over the ACLR protocol. The patient's desired and targeted sports are a key element in the individualization of the rehabilitation programme. Psychosocial-contextual factors should also be integrated to optimize patient management. The planning of the rehabilitation and the RTS decision-making process should take all these requirements into account, in addition to careful consideration of biological, psychological, and biomechanical readiness because each of these factors can contribute to the success or failure of ACLR (Fig. 1).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.rehab.2021.101601](https://doi.org/10.1016/j.rehab.2021.101601).

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